Rockchip UFS Developer Guide

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Preface

Overview

Product Version

Chipset	Kernel Version
RK3576	6.1

Intended Audience

This document (this guide) is mainly intended for:

Technical support engineers

Software development engineers

Revision History

Version	Author	Date	Change Description
V1.0.0	Zhao Yifeng	2024-07-20	Initial version
V1.1.0	Lin Tao	2024-08-09	Supplement configuration and issues
V1.2.0	Lin Tao	2024-08-16	Supplement mirror making exceptions
V1.3.0	Zhao Yifeng	2024-08-20	Add MPHY power supply test configuration Correct some erroneous descriptions
V1.4.0	Lin Tao	2024-08-21	Supplement common UFS device information query
V1.5.0	Lin Tao	2024-09-10	Supplement dual storage uboot detection failure description
V1.6.0	Lin Tao	2025-12-17	Supplementary write speed degradation after drive initialization.

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1. Chip Resource Introduction

RK3576

Resource	Compatible UFS Versions	PHY Supported Speeds	PHY Power Supply	Remarks
UFS	2.1, 2.2, 3.0, and 3.1	HS G1-G3 and PWM G1-G3	0.85v and 1.8v	

Precautions:

1. Do not initialize UFS without PHY power supply; the system will hang.
2. For devices not using UFS, configure the UFS node as "disabled" in the DTS; otherwise, it will affect boot speed.

2. DTS Configuration

Resource	Reference Configuration	Controller Node	PHY Node
UFS	rk3576.dtsi	ufs: ufs@2a2d0000	--

1. `compatible = "rockchip,rk3576-ufs";`

Mandatory Configuration: Default configuration, corresponding driver: drivers/ufs/host/ufs-rockchip.c.

2. `assigned-clock-parents = <&cru CLK_REF_MPHY_26M>`

Mandatory Configuration: Default selection of PPLL output 26MHz reference clock.

If an external independent 26Mhz crystal oscillator is used, it can be configured as: `assigned-clock-parents = <&clk_gpio_mphy_i>;`

Additionally, the dts also needs to specify the clock and frequency of the IO input:

```
clk_gpio_mphy_i: clk_gpio_mphy_i{
            compatible = "fixed-clock";
            #clock-cells = <0>;
            clock-frequency = <26000000>;
            clock-output-names = "clk_gpio_mphy_i";
};
```

3. `status = <okay>;`

Mandatory Configuration: Enable UFS. If the project does not use UFS, it is recommended to configure it as "disabled", otherwise, it will affect the boot-up speed.

4. `pinctrl-0 = <&ufs_refclk>;`

Mandatory Configuration: Configure UFS_REFCLK as a functional IO, outputting the reference clock to the UFS chip.

5. `reset-gpios = <&gpio4 RK_PD0 GPIO_ACTIVE_HIGH>;`

Mandatory Configuration: Configure UFS_RSTN as a GPIO, controlling the reset of the UFS chip within the driver.

6. `vcc-supply = <&vcc_ufs_s0>;`

Optional Configuration: Configure the main power supply for the UFS chip.

7. `vccq-supply = <&vcc1v2_ufs_vccq_s0>;`

Optional Configuration: Configure the UFS VCCQ power supply.

8. `vccq2-supply = <&vcc1v8_ufs_vccq2_s0>;`

Optional Configuration: Configure the UFS VCCQ2 power supply.

9. `freq-table-hz = <50000000 250000000>, <0 0>, <0 0>, <0 0>;`

Optional Configuration: Configure the UFS dvfs frequency adjustment intervals. Each array element defines the minimum and maximum clocks, corresponding one-to-one with the clocks referenced by the UFS node in the dtsi. When there is no IO transmission, the frequency of each clock route will drop to the lowest, and it will rise to the highest before transmission starts. If a route does not require dynamic adjustment, set its maximum and minimum values to 0.

3. Menuconfig Configuration

Ensure the following configurations are enabled.

```
CONFIG_SCSI_UFSHCD=y
CONFIG_SCSI_UFS_BSG=y
CONFIG_SCSI_UFS_HWMON=y
CONFIG_SCSI_UFSHCD_PLATFORM=y
CONFIG_SCSI_UFS_ROCKCHIP=y
```

4. U-Boot Support for UFS

The default SDK code already supports UFS, and direct compilation will enable UFS support.

4.1 DTS Modification

The configuration of DTS nodes is similar to the kernel, and it is necessary to enable the configuration in rk3576-u-boot.dtsi under u-boot.

```
&ufs {
    u-boot,dm-spl;
    status = "okay";
};
```

4.2 Configuration for UFS

UFS requires SCSI protocol support, and the SCSI protocol must be configured.

```
CONFIG_SPL_UFS_SUPPORT=y
CONFIG_CMD_UFS=y
CONFIG_UFS=y
CONFIG_ROCKCHIP_UFS=y
CONFIG_SCSI=y
CONFIG_DM_SCSI=y
```

4.3 UFS MPHY Power Supply Detection

Accessing the UFS MPHY registers without power supply can cause the system to hang. The usbplug and uboot will first detect whether the UFS MPHY is powered before initializing the UFS.

However, the detection code is not compatible with some older UFS chips. The uboot code provides a configuration option to disable detection, refer to the patch below:

```
diff --git a/configs/rk3576-usbplug.config b/configs/rk3576-usbplug.config
```

```
index 822967e7cf5..26cfa5956fe 100644
--- a/configs/rk3576-usbplug.config
+++ b/configs/rk3576-usbplug.config
@@ -101,3 +101,4 @@ CONFIG_SCSI=y
 CONFIG_CMD_UFS=y
 CONFIG_ROCKCHIP_UFS=y
 CONFIG_CMD_SCSI=y
+CONFIG_ROCKCHIP_UFS_DISABLED_LINKUP_TEST=y
diff --git a/configs/rk3576_defconfig b/configs/rk3576_defconfig
index 3ec1fdefe00..15d0451ddc9 100644
--- a/configs/rk3576_defconfig
+++ b/configs/rk3576_defconfig
@@ -223,3 +223,4 @@ CONFIG_RK_AVB_LIBAVB_USER=y
 CONFIG_OPTEE_CLIENT=y
 CONFIG_OPTEE_V2=y
 CONFIG_OPTEE_ALWAYS_USE_SECURITY_PARTITION=y
+CONFIG_ROCKCHIP_UFS_DISABLED_LINKUP_TEST=y
```

After modification, you need to recompile usbplug and uboot. The compilation method is as follows:

```
#echo Compile usbplug
./make.sh rk3576-usbplug
#echo Copy usbplug.bin to rkbin to replace the corresponding file, the file name below is just for reference
cp usbplug.bin ../rkbin/bin/rk35/rk3576_usbplug_v1.02.bin
#Recompile uboot and loader
./make.sh rk3576
```

5. UFS LUN Customization

UFS requires LUN configuration before it can be used, and the default SDK configures 4 LUNs. For details, see the table below:

LUN ID	Name	Size	FLASH Mode	Supports BOOT	Purpose
0	user (sda)	Total Capacity - 48MB	XLC	No	Store system firmware and user data
1	boot0 (sdb)	4MB	SLC	Yes	Store loader
2	boot1 (sdc)	4MB	SLC	Yes	Store backup loader
3	data (sdd)	8MB	SLC	No	Customizable by user

If the default configuration does not meet the requirements, you can modify the LUN configuration yourself. The specific code is in the uboot project's function `ufs_lu_configuration`.

After modification, you need to recompile usbplug. The compilation method is as follows:

```
#echo Compile usbplug
./make.sh rk3576-usbplug
#echo Copy usbplug.bin to rkbin to replace the corresponding file, the file name
below is just for reference
cp usbplug.bin ../rkbin/bin/rk35/rk3576_usbplug_v1.02.bin
#Recompile uboot and loader
./make.sh rk3576
```

6. Vendor Storage Support

6.1 Uboot

Uboot supports Vendor Storage by default, without any restrictions.

6.2 Kernel

The UFS driver under the Kernel does not support Vendor Storage, but some drivers have read requirements during the kernel phase. In such cases, a "ram-vendor-storage" node and reserved memory can be defined in the dts, and the uboot will pass the read data to the kernel driver for use.

```
diff --git a/arch/arm64/boot/dts/rockchip/rk3576-android.dtsi
b/arch/arm64/boot/dts/rockchip/rk3576-android.dtsi
index 4e6574156dc0..b4ec015c2638 100644
--- a/arch/arm64/boot/dts/rockchip/rk3576-android.dtsi
+++ b/arch/arm64/boot/dts/rockchip/rk3576-android.dtsi
@@ -55,6 +55,17 @@ ramoops: ramoops@40110000 {
                        ftrace-size = <0x00000>;
                        record-size = <0x14000>;
                };
+
+               vendor_storage_rm: vendor-storage-rm@00000000 {
+                       compatible = "rockchip,vendor-storage-rm";
+                       reg = <0x0 0x0 0x0 0x0>;
+               };
        };
+
+       vendor_storage: vendor-storage {
+               compatible = "rockchip,ram-vendor-storage";
+               memory-region = <&vendor_storage_rm>;
+               status = "okay";
+       };
 };
```

For writing to vendor storage at the application layer, one can refer to the code in "hardware/rockchip/libvendor_storage/vendor_storage_test.c" for reading and writing operations.

Writing to vendor storage at the kernel level requires the data to be written through an application layer proxy, which is quite cumbersome. The write interface provided by the kernel is for writing to memory and will not actually be saved to the UFS storage.

7. UFS Interface Performance

Using FIO to directly read and write to the sda device without additional file system and encryption/decryption overhead, testing the UFS as a 512GB 3.1 version.

7.1 Sequential Write Performance

Test Command:

```
/data/fio -filename=/dev/block/sda -direct=1 -iodepth 32 -thread -rw=write  -
bs=1024k -size=1G -numjobs=8 -runtime=180 -group_reporting -
name=seq_100write_1024k
```

Test Results:

```
seq_100write_1024k: (g=0): rw=write, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-
1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=32
...
fio-2.20
Starting 8 threads
Jobs: 7 (f=1): [W(1),f(3),_(1),f(3)][100.0%][r=0KiB/s,w=966MiB/s][r=0,w=965
IOPS][eta 00m:00s]
seq_100read_1024k: (groupid=0, jobs=8): err= 0: pid=2389: Wed Jun 26 16:34:04
2024
  write: IOPS=946, BW=947MiB/s (993MB/s)(8192MiB/8652msec)
    clat (msec): min=3, max=32, avg= 8.25, stdev= 2.53
     lat (msec): min=3, max=32, avg= 8.42, stdev= 2.53
    clat percentiles (usec):
     |  1.00th=[ 7072],  5.00th=[ 7136], 10.00th=[ 7200], 20.00th=[ 7264],
     | 30.00th=[ 7264], 40.00th=[ 7264], 50.00th=[ 7264], 60.00th=[ 7328],
     | 70.00th=[ 7392], 80.00th=[ 9536], 90.00th=[ 9920], 95.00th=[14400],
     | 99.00th=[14912], 99.50th=[25728], 99.90th=[29824], 99.95th=[32384],
     | 99.99th=[32384]
   bw (  KiB/s): min=89495, max=126959, per=0.01%, avg=121531.16, stdev=7781.67
    lat (msec) : 4=0.02%, 10=90.32%, 20=9.12%, 50=0.54%
  cpu          : usr=2.04%, sys=5.14%, ctx=16467, majf=0, minf=0
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwt: total=0,8192,0, short=0,0,0, dropped=0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=32

Run status group 0 (all jobs):
  WRITE: bw=947MiB/s (993MB/s), 947MiB/s-947MiB/s (993MB/s-993MB/s), io=8192MiB
(8590MB), run=8652-8652msec

Disk stats (read/write):
  sda: ios=0/16382, merge=0/0, ticks=0/126485, in_queue=126485, util=98.29%
```

7.2 Sequential Read Performance

Test Command:

```
/data/fio -filename=/dev/block/sda -direct=1 -iodepth 32 -thread -rw=read  -
bs=1024k -size=1G -numjobs=8 -runtime=180 -group_reporting -
name=seq_100read_1024k
```

Test Results:

```
seq_100read_1024k: (g=0): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB,
(T) 1024KiB-1024KiB, ioengine=psync, iodepth=32
...
fio-2.20
Starting 8 threads
Jobs: 8 (f=8): [R(8)][100.0%][r=1020MiB/s,w=0KiB/s][r=1020,w=0 IOPS][eta
00m:00s]
seq_100read_1024k: (groupid=0, jobs=8): err= 0: pid=2368: Wed Jun 26 16:31:59
2024
   read: IOPS=1001, BW=1002MiB/s (1050MB/s)(8192MiB/8177msec)
    clat (msec): min=5, max=32, avg= 7.96, stdev= 1.69
     lat (msec): min=5, max=33, avg= 7.96, stdev= 1.69
    clat percentiles (usec):
     |  1.00th=[ 7584],  5.00th=[ 7712], 10.00th=[ 7712], 20.00th=[ 7776],
     | 30.00th=[ 7840], 40.00th=[ 7840], 50.00th=[ 7840], 60.00th=[ 7840],
     | 70.00th=[ 7840], 80.00th=[ 7904], 90.00th=[ 7968], 95.00th=[ 7968],
     | 99.00th=[ 8160], 99.50th=[20096], 99.90th=[33024], 99.95th=[33024],
     | 99.99th=[33024]
   bw (  KiB/s): min=96256, max=132395, per=0.01%, avg=128637.07, stdev=8204.70
    lat (msec) : 10=99.44%, 20=0.06%, 50=0.50%
  cpu          : usr=0.38%, sys=3.85%, ctx=24282, majf=0, minf=2048
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwt: total=8192,0,0, short=0,0,0, dropped=0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=32

Run status group 0 (all jobs):
   READ: bw=1002MiB/s (1050MB/s), 1002MiB/s-1002MiB/s (1050MB/s-1050MB/s),
io=8192MiB (8590MB), run=8177-8177msec

Disk stats (read/write):
  sda: ios=16384/0, merge=0/0, ticks=119146/0, in_queue=119145, util=98.96%
```

7.3 Random Write Performance

Test Command:

```
/data/fio -filename=/dev/block/sda -direct=1 -iodepth 32 -thread -rw=randwrite
 -bs=4k -size=1G -numjobs=8 -runtime=180 -group_reporting -name=rand_100write_4k
```

Test Results:

```
rand_100write_4k: (g=0): rw=randwrite, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T)
4096B-4096B, ioengine=psync, iodepth=32
```

```
...
fio-2.20
Starting 8 threads
Jobs: 7 (f=7): [_(1),w(7)][100.0%][r=0KiB/s,w=270MiB/s][r=0,w=69.0k IOPS][eta
00m:00s]
seq_100read_1024k: (groupid=0, jobs=8): err= 0: pid=2402: Wed Jun 26 16:35:55
2024
  write: IOPS=55.2k, BW=215MiB/s (226MB/s)(8192MiB/38022msec)
    clat (usec): min=35, max=18933, avg=137.53, stdev=421.10
     lat (usec): min=36, max=18935, avg=138.68, stdev=421.21
    clat percentiles (usec):
     |  1.00th=[   49],  5.00th=[   56], 10.00th=[   61], 20.00th=[   68],
     | 30.00th=[   75], 40.00th=[   82], 50.00th=[   88], 60.00th=[   96],
     | 70.00th=[  108], 80.00th=[  127], 90.00th=[  183], 95.00th=[  278],
     | 99.00th=[  532], 99.50th=[  796], 99.90th=[ 6688], 99.95th=[ 6944],
     | 99.99th=[11584]
   bw (  KiB/s): min=22220, max=39863, per=0.01%, avg=27702.29, stdev=2097.44
    lat (usec) : 50=1.25%, 100=62.00%, 250=30.71%, 500=4.86%, 750=0.66%
    lat (usec) : 1000=0.07%
    lat (msec) : 2=0.03%, 4=0.01%, 10=0.40%, 20=0.02%
  cpu          : usr=3.73%, sys=13.05%, ctx=3190681, majf=0, minf=0
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwt: total=0,2097152,0, short=0,0,0, dropped=0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=32

Run status group 0 (all jobs):
  WRITE: bw=215MiB/s (226MB/s), 215MiB/s-215MiB/s (226MB/s-226MB/s), io=8192MiB
(8590MB), run=38022-38022msec

Disk stats (read/write):
  sda: ios=0/2096660, merge=0/13, ticks=0/180540, in_queue=180540, util=99.93%
```

7.4 Random Read Performance

Test Command:

```
/data/fio -filename=/dev/block/sda -direct=1 -iodepth 32 -thread -rw=randread  -
bs=4k -size=1G -numjobs=8 -runtime=180 -group_reporting -name=rand_100read_4k
```

Test Results:

```
rand_100read_4k: (g=0): rw=randread, bs=(R) 4096B-4096B, (W) 4096B-4096B, (T)
4096B-4096B, ioengine=psync, iodepth=32
...
fio-2.20
Starting 8 threads
[ 1393.016346][  T542] healthd: battery l=50 v=3300 t=2.6 h=2 st=3 c=-1600
fc=100 chg=au
Jobs: 8 (f=8): [r(8)][100.0%][r=223MiB/s,w=0KiB/s][r=56.0k,w=0 IOPS][eta
00m:00s]
seq_100read_1024k: (groupid=0, jobs=8): err= 0: pid=2412: Wed Jun 26 16:37:30
2024
```

```
   read: IOPS=57.5k, BW=224MiB/s (235MB/s)(8192MiB/36499msec)
    clat (usec): min=60, max=2949, avg=133.28, stdev=34.67
     lat (usec): min=61, max=2949, avg=133.69, stdev=34.78
    clat percentiles (usec):
     |  1.00th=[   95],  5.00th=[   99], 10.00th=[  103], 20.00th=[  108],
     | 30.00th=[  113], 40.00th=[  118], 50.00th=[  124], 60.00th=[  131],
     | 70.00th=[  141], 80.00th=[  153], 90.00th=[  173], 95.00th=[  197],
     | 99.00th=[  262], 99.50th=[  294], 99.90th=[  374], 99.95th=[  418],
     | 99.99th=[  580]
   bw (  KiB/s): min=27040, max=29907, per=0.01%, avg=28834.42, stdev=390.23
    lat (usec) : 100=5.06%, 250=93.56%, 500=1.36%, 750=0.02%, 1000=0.01%
    lat (msec) : 2=0.01%, 4=0.01%
  cpu          : usr=3.61%, sys=13.20%, ctx=3825319, majf=0, minf=8
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
     submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
     issued rwt: total=2097152,0,0, short=0,0,0, dropped=0,0,0
     latency   : target=0, window=0, percentile=100.00%, depth=32

Run status group 0 (all jobs):
   READ: bw=224MiB/s (235MB/s), 224MiB/s-224MiB/s (235MB/s-235MB/s), io=8192MiB
(8590MB), run=36499-36499msec

Disk stats (read/write):
  sda: ios=2088272/0, merge=16/0, ticks=222685/0, in_queue=222685, util=99.91%
```

8. UFS Device Information Inquiry and Operations

On the Linux kernel platform, the ufs-utils tool can be used to inquire and operate on the related information of UFS devices. Common commands are as follows for reference:

1. Inquiry of Chip Health Status

```
ufs-utils desc -t 9 -p /dev/bsg/ufs-bsg0  # Linux platform
ufs-utils desc -t 9 -p /dev/ufs-bsg0      # Android platform
Device Health Descriptor: [Byte offset 0x0]: bLength = 0x25
Device Health Descriptor: [Byte offset 0x1]: bDescriptorType = 0x9
Device Health Descriptor: [Byte offset 0x2]: bPreEOLInfo = 0x1
Device Health Descriptor: [Byte offset 0x3]: bDeviceLifeTimeEstA = 0x1
Device Health Descriptor: [Byte offset 0x4]: bDeviceLifeTimeEstB = 0x1

bPreEOLInfo:
Pre End of Life Information
00h: Not defined
01h: Normal
02h: Warning. Consumed 80% of reserved blocks.
03h: Critical. Consumed 90% of reserved blocks.
Others: Reserved

bDeviceLifeTimeEstA
bDeviceLifeTimeEstB:
provides an indication of the device life time based on the amount of performed
program/erase cycles
00h: Information not available
```

```
01h: 0% - 10% device life time used
02h: 10% - 20% device life time used
03h: 20% - 30% device life time used
04h: 30% - 40% device life time used
05h: 40% - 50% device life time used
06h: 50% - 60% device life time used
07h: 60% - 70% device life time used
08h: 70% - 80% device life time used
09h: 80% - 90% device life time used
0Ah: 90% - 100% device life time used
0Bh: Exceeded its maximum estimated device life time
Others: Reserved
```

2. Inquiry of the Current Chip's Running Protocol Version, Speed, and Lane Count

```
# Inquiry of the current protocol version being run by both parties:
ufs-utils desc -t 9 -p /dev/bsg/ufs-bsg0  # Linux platform
ufs-utils desc -t 9 -p /dev/ufs-bsg0      # Android platform

# Inquiry of link status
ufs-utils uic -t 1 -a -p /dev/bsg/ufs-bsg0 # Linux platform
ufs-utils uic -t 1 -a -p /dev/ufs-bsg0     # Android platform

1. Speed (For HS Gear interface speed, please refer to the appendix):
Gear: 1: Gear1  2: Gear2  3: Gear3  4: Gear4
HSSeries: 1: rate A  2: rate B
[0x1568]PA_TxGear               : local = 0x00000001, peer = 0x00000001
[0x1583]PA_RxGear               : local = 0x00000001, peer = 0x00000001
[0x156a]PA_HSSeries             : local = 0x00000002, peer = 0x00000002

2. Lane Count:
The number of physical layer lanes that both parties can provide
[0x1520]PA_AvailTxDataLanes     : local = 0x00000002, peer = 0x00000002
[0x1540]PA_AvailRxDataLanes     : local = 0x00000002, peer = 0x00000002
The number of physical layer lanes that both parties are actually linked on
[0x1561]PA_ConnectedTxDataLanes  : local = 0x00000002, peer = 0x00000002
[0x1581]PA_ConnectedRxDataLanes  : local = 0x00000002, peer = 0x00000002
The number of lanes that both parties are actually using for communication
[0x1560]PA_ActiveTxDataLanes     : local = 0x00000001, peer = 0x00000001
[0x1580]PA_ActiveRxDataLanes     : local = 0x00000001, peer = 0x00000001

If the number of lanes that both parties are actually linked on is less than the
number of physical layer lanes that both parties can provide, it indicates that
poor signal quality has caused some lanes to fail training, or that the actual
hardware only has one lane connected.
If the number of lanes that both parties are actually using for communication is
less than the number of lanes that both parties are actually linked on, it
indicates that the configuration information has been modified, possibly by the
ufs driver modifying pwr_info or by modifying through ufs-utils.
```

9. Common Issues

9.1 Failure to Boot After Firmware Upgrade, GPT Error

The block size of UFS is 4K, and GPT needs to be generated according to the 4K block size. The tools need to be updated to the following corresponding versions or newer versions:

```
RKDevTool_v3.28_for_window

FactoryTool_v1.88

upgrade_tool_v2.30_for_linux
```

9.2 Firmware Upgrade Failure and Lack of Serial Output

The "BOOT MODE CONFIG" setting is incorrect; refer to the "BOOT MODE CONFIG" table in the circuit schematic for UFS boot configuration.

An unsupported UFS chip is being used; try replacing it with a brand-new UFS chip of a different model.

9.3 Monitoring Exception Information During Kernel Aging Process

```
bash-5.2# cat /sys/kernel/debug/ufshcd/2a2d0000.ufs/stats
PHY Adapter Layer errors (except LINERESET): 0  # UECPA type
Data Link Layer errors: 0                       # UECDL type
Network Layer errors: 0                         # UECN type
Transport Layer errors: 0                       # UECT type
Generic DME errors: 0                           # UECDME type
Auto-hibernate errors: 0                        # UHES and UHXS types
IS Fatal errors (CEFES, SBFES, HCFES, DFES): 0
DME Link Startup errors: 0                      # DME link startup errors
PM Resume errors: 0                             # Wake-up failures
PM Suspend errors : 0                           # Standby failures
Logical Unit Resets: 1                          # Number of times the device is
reset, increased by booting, waking from hibernation, and error handling
Host Resets: 0                                  # Number of times the controller
is reset and re-initialized, increased by errors
SCSI command aborts: 0                          # Number of times SCSI command
communication errors occur
```

9.4 Kernel Failure to Mount Partitions Post-Boot

```
[    3.220417]  sda: sda1 sda2 sda3 sda4 sda5 sda6 sda7 sda8 sda9
[    3.221108] sd 0:0:0:0: [sda] Attached SCSI disk
[    3.467971] EXT4-fs (sda7): bad block size 1024
[    3.468341] EXT4-fs (sda7): bad block size 1024
[    3.468528] EXT4-fs (sda7): bad block size 1024
[    3.468733] FAT-fs (sda7): utf8 is not a recommended IO charset for FAT
filesystems, filesystem will be case sensitive!
[    3.468903] ISOFS: unsupported/invalid hardware sector size 4096
[    3.470737] EXT4-fs (sda7): bad block size 1024
[    3.471013] EXT4-fs (sda7): bad block size 1024
[    3.471260] EXT4-fs (sda7): bad block size 1024
```

```
[    3.471494] FAT-fs (sda7): utf8 is not a recommended IO charset for FAT
filesystems, filesystem will be case sensitive!
[    3.471659] ISOFS: unsupported/invalid hardware sector size 4096
[    3.472991] List of all partitions:
[    3.473009] 0100            4096 ram0
[    3.473014]  (driver?)
[    3.473035] 0800       124936192 sda
[    3.473039]  driver: sd
[    3.473050]   0801            8192 sda1 17b18c43-1d24-4484-8883-d0c618e42411
[    3.473054]
[    3.473065]   0802            8192 sda2 97c64638-8824-46d3-d83e-582a6a48d4fb
[    3.473070]
[    3.473080]   0803            4096 sda3 e6edc612-f94f-44b7-b9f1-1561474fafbb
[    3.473084]
[    3.473095]   0804           65536 sda4 d04a9010-9d09-4d47-96ba-3fa854d3c5c5
[    3.473099]
[    3.473110]   0805          131072 sda5 f2d9eb7f-2700-4ba8-9422-0e502db7c35c
[    3.473114]
[    3.473124]   0806           32768 sda6 1382357d-a655-48e7-f1da-99e05054ea34
[    3.473129]
[    3.473139]   0807        14680064 sda7 614e0000-0000-4b53-8000-1d28000054a9
[    3.473143]
[    3.473153]   0808          131072 sda8 d5f63744-ff57-4447-c9ee-b98e63e9b69b
[    3.473158]
[    3.473168]   0809       109871084 sda9 ec8c0d6e-d426-462e-bfc0-366c6e064804
[    3.473172]
[    3.473185] 0810            4096 sdb
[    3.473189]  driver: sd
[    3.473201] 0820            4096 sdc
[    3.473205]  driver: sd
[    3.473217] 0830            8192 sdd
```

The cause of the failure is that the partition image was created with an incorrect block size specified by the tool. As shown in the log above, the ext4 image was created with a block size of 1KB, while UFS requires support for a block size of 4KB.

9.5 U-Boot Detection Failure of UFS

When the system supports dual-storage boot, such as eMMC+UFS, and UFS is not the primary storage device, failure in detection may lead to U-Boot disabling support for UFS in the DTB. Consequently, upon entering the kernel, there will be no initialization logs for the UFS controller. This situation typically indicates a UFS hardware issue. Ensure that the chip is supported in our AVL list and that the card has been powered on in maskrom upgrade mode, then proceed to check the chip power supply, peripheral components, etc., according to the reference schematic.

```
Timedout waiting for UIC response
Host controller enable failed
ufshcd_pltfrm_init() failed -5
FDT: UFS was not detected, disabling UFS.
```

9.6 Write performance degradation after re-initialization

After re-initialization on our platform, some UFS devices particles may exhibit degraded write performance or speeds significantly below the specifications outlined in the datasheet. This occurs because, for products without battery backup, there is a risk of data loss in the event of a power failure. To mitigate this risk, our re-initialization process enables the reliable write mode by default. If necessary, you may disable the reliable write mode, recompile the `usbplug` tool, and perform the initialization again. However, please carefully evaluate the risk of power-loss data corruption based on your specific product's characteristics and consult the original manufacturer for assessment before making this change.

```
--- a/drivers/ufs/ufs-rockchip-usbplug.c
+++ b/drivers/ufs/ufs-rockchip-usbplug.c
        /* lu 0: data lu, max capacity*/
@@ -294,7 +294,7 @@ static void ufs_lu_configuration(struct ufs_hba *hba, struct
ufs_configuration_d
        for (i = 0; i <= 3; i++) { /* lu 0 - 3 */
                unit[i].b_lu_enable = 0x1;
                unit[i].b_lu_write_protect = 0x0;
-               unit[i].b_data_reliability = 0x1;
+               unit[i].b_data_reliability = 0x0;
                unit[i].b_logical_block_size = 0x0c;
                unit[i].b_provisioning_type = 0x2;
                unit[i].w_context_capabilities = 0x0;
```

10. Appendix

10.1 ufs-utils Download Address

After obtaining Redmine permissions, you can directly access https://redmine.rock-chips.com/documents/108 to acquire the ufs-utils program.

10.2 HS Gear Information

HS-GEAR	Rate A-series	Rate B-series		Rate A-series (from [MIPI-M-PHY])	Rate B-series[3] (from [MIPI-M-PHY])	Unit
	f_{ref}	f_{ref}		f_{ref}	f_{ref}	
	19.2 / 26 / 38.4	19.2 / 38.4	26	19.2 / 26 / 38.4		MHz
HS-GEAR1	1248 [2]	1459.2	1456.0	1248	1457.6	Mbps
HS-GEAR2	2496	2918.4	2912.0	2496	2915.2	Mbps
HS-GEAR3	4992	5836.8	5824.0	4992	5830.4	Mbps
HS-GEAR4	9984	11673.6	11648.0	9984	11660.8	Mbps